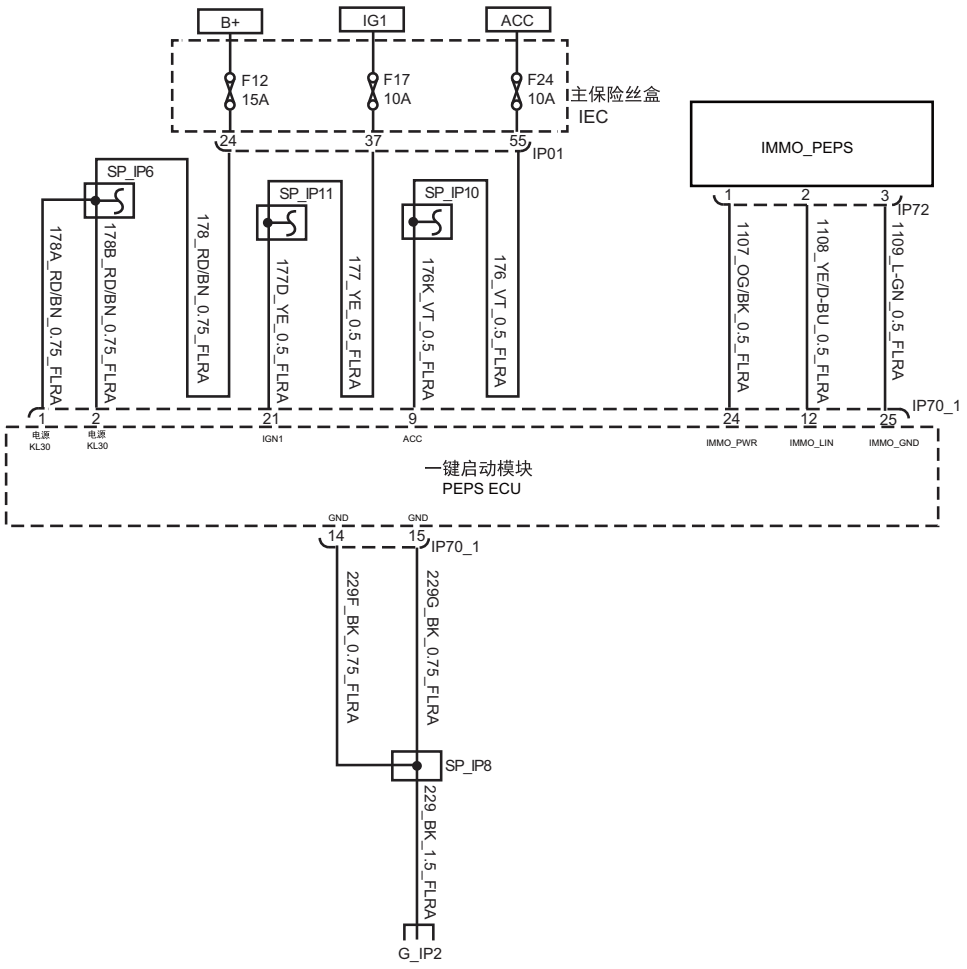
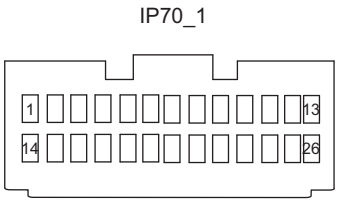
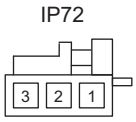
B1501-09

Fault diagnosis code
B1501-09：IMMO antenna eoor
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • BCM Module fault • PEPS Module fault • IMMO antenna error • Insurance fuse fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
B1501-09
IMMO antenna error
To set the effect of a fault code condition
IMMO function fault
Description of circuit diagram

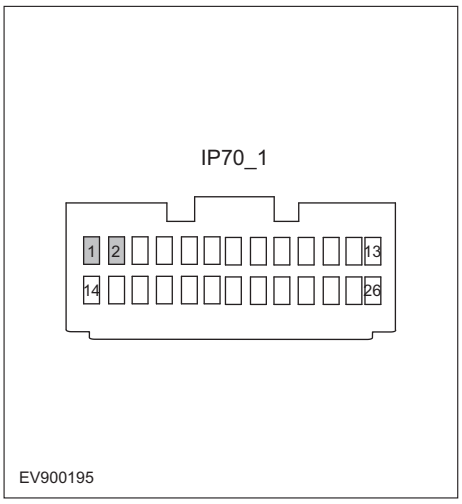
Circuit diagram

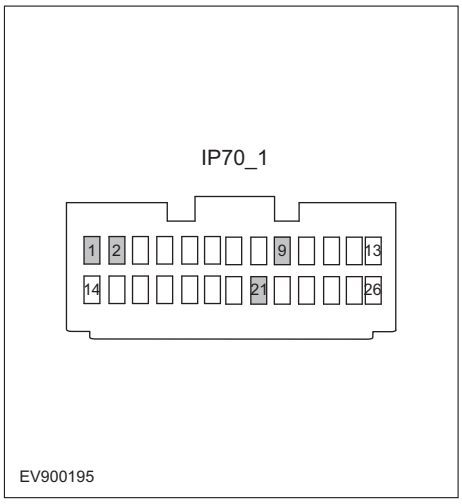
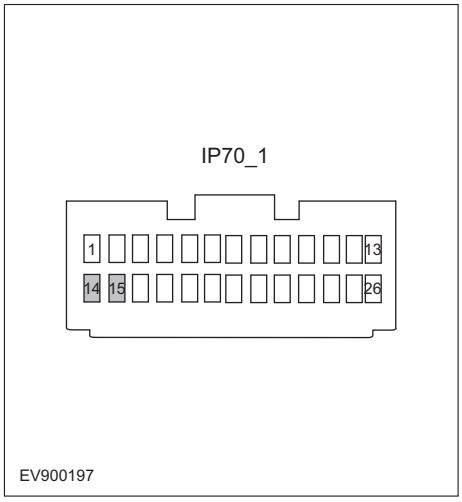


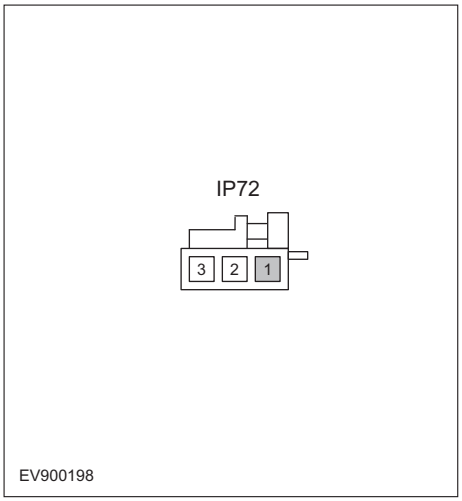


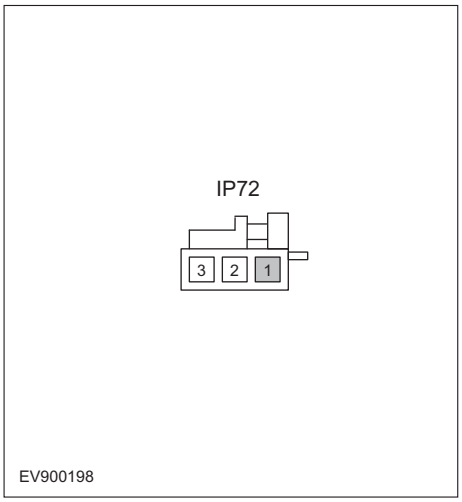


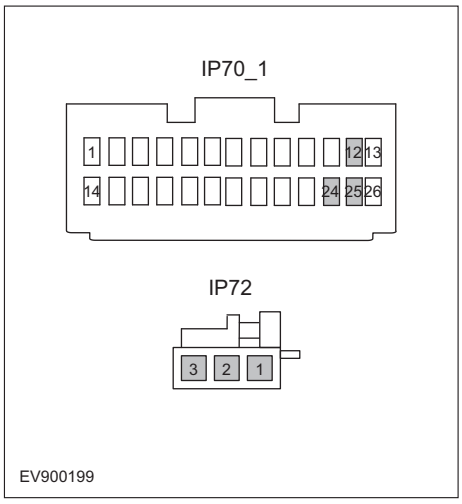
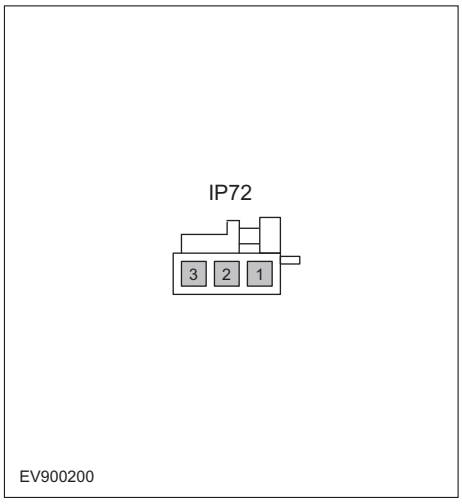
EV900194

Troubleshooting steps	
Step 1	General inspection.
	1 Check F12,F17,F24 fuses for damages. 2 Check battery capacity. 3 Check the harness plug of IMMO, PEPS, BCM for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the BCM with diagnostic apparatus. 4 Is there any other fault code except for B1501-09?
	YES → Refer to:DTC Summary list(BCM).
NO ↓	
Step 3	Check the power voltage of PEPS battery.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP70_1 of PEPS. 3 Measure the voltage between the harness terminal of PEPS and ground. Measuring circuit: voltage between terminal 1 on plug IP70_1 and ground. Measuring circuit: voltage between terminal 2 on plug IP70_1 and ground. **Standard value:10~14V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 4	Check the power voltage when the PEPS is started or operating
	1 Turn the ignition switch to ON. 2 Measure the voltage between the harness terminal of PEPS and ground. Measuring circuit: voltage between terminal 1 on plug IP70_1 and ground. Measuring circuit: voltage between terminal 2 on plug IP70_1 and ground. Measuring circuit: voltage between terminal 9 on plug IP70_1 and ground. Measuring circuit: voltage between terminal 21 on plug IP70_1 and ground. **Standard value:10~14V** 3 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 5	Test whether the grounding circuit of PEPS is open.
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness terminal of PEPS and ground. Measuring circuit: resistance between terminal 14 on plug IP70_1 and ground. Measuring circuit: resistance between terminal 15 on plug IP70_1 and ground. **Standard value: < 1Ω** 3 Test whether the resistance is less than 1Ω?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the power voltage of IMMO antenna battery.
	1 Turn the ignition switch to OFF. 2 Connect the harness plug IP70_1 of PEPS. 3 Disconnect the harness plug IP72 of IMMO antenna. 4 Measure the voltage between the harness terminal of IMMO antenna and ground. Measuring circuit: voltage between terminal 1 on plug IP72 and ground. **Standard value:10~14V** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 7	Check the power voltage when the IMMO antenna is started or operating
	1 Turn the ignition switch to ON. 2 Measure the voltage between the harness terminal of IMMO antenna and ground. Measuring circuit: voltage between terminal 1 on plug IP72 and ground. **Standard value:10~14V** 3 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 8	Check the circuit between IMMO antenna and PEPS (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP70_1 of PEPS. 3 Measure the resistance between the harness terminal of IMMO antenna and the PEPS. Measuring circuit: resistance between terminal 1 on plug IP72 and terminal 24 on plug IP70_1. Measuring circuit: resistance between terminal 2 on plug IP72 and terminal 12 on plug IP70_1. Measuring circuit: resistance between terminal 2 on plug IP72 and terminal 25 on plug IP70_1. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 9	Check the circuit of IMMO antenna(short circuit).
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness terminal of IMMO antenna and ground. Measuring circuit: resistance between terminal 1 on plug IP72 and ground. Measuring circuit: resistance between terminal 2 on plug IP72 and ground. Measuring circuit: resistance between terminal 3 on plug IP72 and ground. **Standard value: ∞** 3 Measure the voltage between the harness terminal of IMMO antenna and ground. Measuring circuit: voltage between terminal 1 on plug IP72 and ground. Measuring circuit: voltage between terminal 2 on plug IP72 and ground. Measuring circuit: voltage between terminal 3 on plug IP72 and ground. **Standard value: ≈ 0V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 10	Test whether BCM is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB09_C、EB09_D、IP09_A、IP09_B of BCM. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BCM plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Change the BCM Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	